Consumer Behavior Analysis Report

Leveraging Shopping Data to Identify Trends, Improve Customer Engagement, and Optimize Strategy

Prepared by

Hamza Idrissi Meliani

Mathematics and Systems Engineering Student

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Tools Used: **Python (Pandas) • MySQL (SQL) • Power BI**

Dataset: **3,900 customer records • 18 variables**

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1. Executive Summary

This report presents a comprehensive analysis of a retail company's customer shopping behavior dataset comprising **3,900 customer records** and **18 variables**. The analysis was conducted using a full data pipeline — Python for data preparation, MySQL for structured querying, and Power BI for interactive visualization.

The central business question addressed is:

> *"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"*

Key findings at a glance:

- Total revenue across all customers: **$233,081**, with male customers contributing approximately **67.7%** of total revenue.
- Average purchase amount: **$59.76**; average review rating: **3.75 / 5**.
- **Clothing** is the dominant category, accounting for roughly **44.7%** of total revenue.
- Only **27%** of customers hold an active subscription, yet repeat buyers (>5 purchases) show a **23% higher** subscription rate than non-repeat buyers.
- **Fall** is the highest-revenue season; **Young Adults** are the top-spending age group.
- **79.6%** of customers are classified as *Loyal* based on their purchase history.

2. Problem Statement

A leading retail company seeks to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has observed shifts in purchasing patterns across:

- **Demographics** — age groups and gender
- **Product categories** — clothing, accessories, footwear, and outerwear
- **Sales channels** — online vs. offline, shipping preferences

They are particularly interested in uncovering which factors — such as discounts, review ratings, seasonal trends, and payment preferences — drive consumer decisions and repeat purchases.

3. Project Scope & Deliverables

#	Deliverable	Description
1	**Data Preparation & Modeling (Python)**	Clean and transform the raw dataset: handle missing values, rename columns, engineer new features (`age_group`, `purchase_frequency_days`), and remove redundant fields.
2	**Data Analysis (SQL)**	Organize data in MySQL, simulate business transactions, and run 10 analytical queries covering revenue, loyalty, discounts, and segmentation.
3	**Visualization & Insights (Power BI)**	Interactive dashboard displaying KPIs, revenue by category and age group, subscription breakdown, and sales filters.

Table 1: Project deliverables overview

4. Dataset Overview

4.1. Dataset Description

The dataset (`customer_shopping_behavior.csv`) contains **3,900 rows** and **18 columns** capturing diverse aspects of customer transactions.

Column	Type	Description
customer_id	Integer	Unique customer identifier
age	Integer	Customer age (18–70)
gender	Categorical	Male / Female
item_purchased	Categorical	Product name
category	Categorical	Clothing, Accessories, Footwear, Outerwear
purchase_amount (USD)	Float	Transaction value
location	Categorical	US state
season	Categorical	Spring, Summer, Fall, Winter
review_rating	Float	1–5 star rating (37 missing)
subscription_status	Categorical	Yes / No
shipping_type	Categorical	6 shipping options
discount_applied	Categorical	Yes / No
previous_purchases	Integer	Number of prior orders
payment_method	Categorical	Card, Cash, PayPal, etc.
frequency_of_purchases	Categorical	Weekly, Monthly, etc.

Table 2: Dataset column reference

5. Data Preparation & Modeling (Python)

5.1. Missing Value Treatment

Upon initial inspection, the `Review Rating` column contained **37 missing values** (approximately 0.95% of the dataset). Rather than dropping these rows or applying a global median, a more granular imputation strategy was adopted:

Code

```
df['Review Rating'] = df.groupby('Category')['Review Rating']
.transform(lambda x:  x.fillna(x.median()))
```

The table below shows the null value count **before** and **after** imputation, confirming the fix was successful:

Before Imputation

```
Customer ID                 0
Age                         0
Gender                      0
Item Purchased              0
Category                    0
Purchase Amount (USD)       0
Location                    0
Size                        0
Color                       0
Season                      0
Review Rating              37
Subscription Status         0
Shipping Type               0
Discount Applied            0
Promo Code Used             0
Previous Purchases          0
Payment Method              0
Frequency of Purchases      0
dtype:  int64
```

After Imputation

```
Customer ID                 0
Age                         0
Gender                      0
Item Purchased              0
Category                    0
Purchase Amount (USD)       0
Location                    0
Size                        0
Color                       0
Season                      0
Review Rating               0
Subscription Status         0
Shipping Type               0
Discount Applied            0
Promo Code Used             0
Previous Purchases          0
Payment Method              0
Frequency of Purchases      0
dtype:  int64
```

Figure 1: `df.isnull().sum()` — before and after imputation (cell [38] vs [40])

5.2. Column Standardization

All column names were normalized to `snake_case` for consistency with the SQL database schema. The column `purchase_amount_(usd)` was renamed to `purchase_amount` for cleaner references. The resulting columns after renaming are shown below:

Output — df.columns (cell [42])

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
'purchase_amount', 'location', 'size', 'color', 'season',
'review_rating', 'subscription_status', 'shipping_type',
'discount_applied', 'promo_code_used', 'previous_purchases',
'payment_method', 'frequency_of_purchases'],
dtype='str')
```

5.3. Feature Engineering

Two new features were created to enrich the analytical capabilities of the dataset:

5.3.1. Age Group (`age_group`)

Customers were divided into four equal-sized quartile-based age segments using `pd.qcut()`. The output below (cell [44]) shows the first 10 rows confirming correct assignment:

Output — df[['age','age_group']].head(10) (cell [44])

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

Figure 2: Age group assignment — first 10 records

5.3.2. Purchase Frequency in Days (`purchase_frequency_days`)

The categorical frequency labels were mapped to numerical values (days) to enable quantitative analysis. The output below (cell [46]) confirms correct mapping:

Output — df[['purchase_frequency_days','frequency_of_ (cell [46])

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

Figure 3: Purchase frequency mapping — first 10 records

5.4. Column Removal

The `promo_code_used` column was found to be **100% identical** to `discount_applied`, making it fully redundant. This was first confirmed visually by inspecting both columns side by side (cell [47]), then verified programmatically with a Boolean equality check (cell [48]).

Side-by-side check (cell [47])

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

Boolean equality check (cell [48])

```
(df['discount_applied'] ==
df['promo_code_used']).all()

Output:
np.True_
⇒ Columns are 100% identical.
⇒ promo_code_used dropped.
Features:  18 → 17 columns.
```

Figure 4: Redundancy proof — `promo_code_used` vs `discount_applied`

6. Data Analysis (SQL)

The cleaned DataFrame was exported to a local **MySQL** database (`customer_behavior`) using SQLAlchemy. Ten analytical queries were designed and executed to extract business insights.

6.1. KPI Overview

$233,081	$59.76	3.75 / 5
Total Revenue	Average Purchase Amount	Average Review Rating

6.2. Q1 — Revenue by Gender

Gender	Total Revenue (USD)	Share (%)
Male	$157,890	67.7%
Female	$75,191	32.3%
Total	$233,081	100%

Table 3: Total revenue by gender

Male customers generate more than **twice** the revenue of female customers. This gap warrants investigation: it may reflect a larger male customer base, higher average spend, or a product catalog skewed toward male preferences.

6.3. Q2 — High-Value Discount Users

Customers who applied a discount but still spent above the overall average ($59.76) represent a high-value segment. These customers are simultaneously *deal-conscious* and *high-spending*, making them prime candidates for targeted premium loyalty programs. The query returned **839 customers** matching this profile — a sample of the first 20 records is shown below.

SQL Output — Q2: High-Value Discount Users (sample of 839 results)

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76
43	100
44	69

... 819 more rows

Figure 5: SQL output — Query 2: High-value discount customers (first 20 of 839)

6.4. Q3 — Top 5 Products by Review Rating

Rank	Product	Avg. Rating
1	**Gloves**	3.86
2	**Sandals**	3.84
3	**Boots**	3.82
4	**Hat**	3.80
5	**Skirt**	3.79

Table 4: Top 5 products by average customer review rating

6.5. Q4 — Shipping Type vs. Average Spend

Standard shipping ($58.46) and Express shipping ($60.48) show a modest difference in average purchase amounts. Customers choosing faster shipping options do not necessarily spend significantly more, suggesting shipping choice is more about convenience than cart size.

6.6. Q5 — Subscribers vs. Non-Subscribers

Subscription	Customers	Avg Spend	Total Revenue
No	2,847	$59.80	$170,249
Yes	1,053	$59.65	$62,832

Table 5: Revenue comparison: subscribers vs. non-subscribers

Subscribed customers do *not* spend significantly more per transaction ($59.65 vs $59.80). However, growing the subscriber base remains strategically important for **retention** and **predictable recurring revenue**.

6.7. Q6 — Products with Highest Discount Usage

The five products with the highest discount application rates were identified. Understanding which products are most frequently discounted enables better margin management and targeted promotional strategy.

6.8. Q7 — Customer Loyalty Segmentation

Segment	Rule	Count
Loyal	> 10 previous purchases	3,116 (79.9%)
Returning	2–10 previous purchases	701 (18.0%)
New	1 previous purchase	83 (2.1%)

Table 6: Customer loyalty segmentation

Nearly **80%** of the customer base is classified as *Loyal*. This is a strong retention signal but also suggests limited customer acquisition. The company should investigate strategies to attract new customers.

6.9. Q8 — Top 3 Products per Category

Using a `ROW_NUMBER()` window function partitioned by category, the three most-purchased products in each of the four categories were identified. This is valuable for inventory planning and featured-product marketing.

SQL Output — Q8: Top 3 Products per Category

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

(3rd Outerwear product not shown in result)

Figure 6: SQL output — Query 8: Top 3 most purchased products per category

6.10. Q9 — Repeat Buyers & Subscription Likelihood

Group	Subscription Rate
Repeat Buyers (> 5 purchases)	27.6%
Non-Repeat Buyers (≤ 5 purchases)	22.4%

Table 7: Subscription rate by purchase frequency group

Repeat buyers are **23% more likely** to be subscribed than non-repeat buyers, confirming a positive correlation between engagement depth and subscription adoption.

6.11. Q10 — Revenue by Age Group

Age Group	Total Revenue	Share
Young Adult	$62,143	26.7%
Middle-aged	$59,197	25.4%
Adult	$55,978	24.0%
Senior	$55,763	23.9%

Table 8: Revenue contribution by age group

7. Visualization & Insights (Power BI)

An interactive Power BI dashboard was developed to make the insights accessible to non-technical stakeholders. The dashboard includes slicer filters for **subscription status**, **gender**, **product category**, and **shipping type**.

7.1. Dashboard KPIs

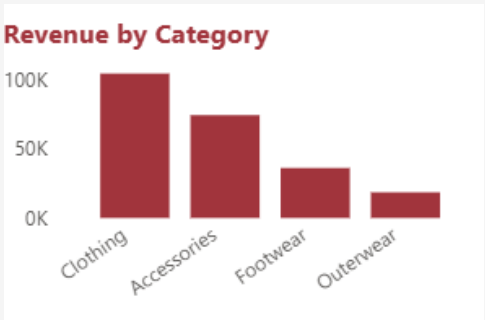
3.9K
Number of Customers

$59.76
Avg Purchase Amount

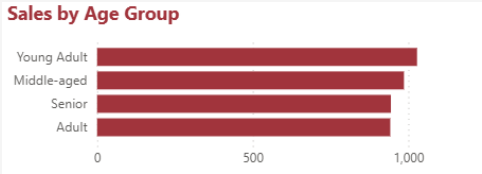
3.75
Avg Review Rating

7.2. Dashboard Screenshots

Figure 7: Customer Behavior Dashboard — Power BI (Page 1)



(a) Revenue by Category



(b) Sales by Age Group

Figure 8: Power BI dashboard visual breakdowns

7.3. Dashboard Features

- **Subscription Donut Chart** — 27% subscribed vs. 73% non-subscribed.
- **Revenue by Category** (bar chart) — Clothing leads, followed by Accessories.
- **Sales by Category** — transaction count across all four categories.
- **Revenue & Sales by Age Group** — horizontal bar charts showing Young Adults as top contributors.
- **Interactive Slicers** — filter by subscription, gender, category, and shipping type.

8. Key Findings

#	Finding	Detail
1	**Gender Revenue Gap**	Male customers account for 67.7% of revenue despite representing a smaller majority of the dataset.
2	**Loyalty Dominance**	79.9% of customers are Loyal (>10 purchases), indicating a highly retained base with low new-customer influx.
3	**Low Subscription Penetration**	Only 27% subscribe, yet subscribers show similar spend per transaction. Subscription growth is a revenue stability lever.
4	**Discount & Spend Correlation**	A notable segment of customers use discounts and still exceed the average spend — a high-value cohort.
5	**Seasonal Peaks**	Fall generates the highest revenue ($60,018), suggesting targeted seasonal campaigns could maximize Q3/Q4 performance.
6	**Young Adults Lead Spending**	Young Adults (18–31) contribute the most revenue ($62,143), making them the most commercially significant demographic.
7	**Clothing Category Dominance**	Clothing accounts for 44.7% of total revenue, solidifying it as the company's core business driver.

Table 9: Summary of key analytical findings

9. Business Recommendations

9.1. Grow the Subscriber Base

With only 27% subscription penetration, there is significant room to grow. Introducing tiered subscription plans with tangible perks (early access, free shipping, exclusive discounts) targeting the 73% non-subscribers would boost both loyalty and revenue predictability.

9.2. Target Female Customers with Tailored Campaigns

Given the revenue gap between male (67.7%) and female (32.3%) customers, the company should analyze whether this reflects a catalog gap or a marketing misalignment. Dedicated female-focused collections and campaigns could unlock a large, under-served segment.

9.3. Leverage the High-Value Discount Cohort

Customers who apply discounts but still spend above average are ideal for a **premium loyalty tier**. Offering them exclusive early-access sales or member-only deals can increase both spend and retention without cannibalizing full-price margins.

9.4. Capitalize on Fall Seasonality

Fall is the top-performing season. The company should pre-plan marketing campaigns, inventory build-up, and promotional pushes for the August–October window to maximize this natural demand spike.

9.5. Invest in New Customer Acquisition

With 79.9% loyal customers, the growth ceiling is approaching. Referral programs, social media acquisition campaigns, and introductory offers should be prioritized to expand the customer base beyond existing loyalists.

9.6. Expand Accessories and Footwear Categories

While Clothing dominates, Accessories ($74,200) and Footwear ($36,093) show solid potential. Expanding product range and visibility in these categories could diversify revenue and reduce over-reliance on Clothing.

10. Methodology & Tools

Phase	Tool	Purpose
Data Preparation	Python (Pandas)	Cleaning, imputation, feature engineering, export to SQL
Database Layer	MySQL + SQLAlchemy	Structured storage and 10 analytical SQL queries
Visualization	Microsoft Power BI	Interactive dashboard with slicers and KPIs

Table 10: Tools used across the project pipeline

11. Conclusion

This end-to-end data analysis project demonstrates that the company possesses a loyal, engaged customer base with measurable purchasing patterns across demographics, categories, and seasons. By structuring data in a relational database, querying it systematically, and visualizing results in an interactive Power BI dashboard, actionable insights were extracted to directly answer the central business question.

The key strategic levers identified are: **subscription growth**, **gender-targeted marketing**, **seasonal campaign optimization**, and **new customer acquisition**. Implementing these recommendations, grounded in data, positions the company to improve both top-line revenue and long-term customer lifetime value.

> *"Data is only as valuable as the decisions it drives. This analysis provides the foundation — the next step is execution."*